

Core 4 — Project Materials & Repository Checklist

In-Memory Acceleration of Monte Carlo and ML-Based Financial Workloads
Chips & AI Hackathon • Theme 4: In-Memory Computing and Circuits

1. What This Project Needs

This is a software/simulation-based proof of concept — no physical hardware, lab equipment, or chip fabrication is required. Everything below is software.

Core Software Stack

Tool / Library	Used for
Python 3.x	Base language for the entire pipeline
NumPy / SciPy	Monte Carlo simulation (GBM paths, random sampling)
Pandas	Building and passing the CSV dataset between team members
PyTorch or TensorFlow/Keras	Neural-network surrogate (5→64→64→1 architecture)
scikit-learn	Train/val/test splits, MAE / RMSE / R-squared metrics
Matplotlib / Seaborn	Accuracy, latency, data-movement, and energy plots
Jupyter	Interactive notebooks for MC validation and NN training

Role-Specific Tools

- **Raya (IMC mapping):** No special simulator needed — NumPy math on the NN's weight matrices, plus a shared doc/spreadsheet to record MAC counts, memory-access, and latency/energy assumptions.
- **Ishika (Benchmarking):** Python's `time/timeit` for runtime measurement, plus the shared results schema everyone writes into.

Collaboration / Infrastructure

- A shared GitHub repo (per-owner folders or per-owner repos)
- A shared location for dataset CSVs, so teammates aren't blocked waiting on final data
- Screen recording / editing software for the final demo video (due Sep 19, 2026)

2. What the Git Repository Should Have

Checklist for a complete project repository, covering both the shared team scaffold and each individual component repo.

Top-Level Structure

- **monte_carlo/** — Nini's MC pricer, Black-Scholes validation, dataset generation script
- **neural_network/** — Rupsaa's NN preprocessing, training, evaluation code
- **imc_mapping/** — Raya's IMC modeling, MAC-count and latency/energy estimation code
- **benchmarking/** — Ishika's benchmark schema, results aggregation, comparison plots
- **data/** — shared CSV datasets (dummy and final)
- **results/** — final tables, figures, frozen results
- **docs/** — project plan, one-page proposal, schedule, meeting notes

Files Every Component Repo Should Have

- **README.md** — what the component does, method/formulas used, actual validation results, dataset schema, how to run it, and what it hands off downstream
- **requirements.txt** — exact dependencies (numpy, pandas, torch, etc.)
- **.gitignore** — excludes `__pycache__/, .ipynb_checkpoints/`, virtual envs, and large generated files that don't need version control
- A **.py module** version of notebook logic where practical — reusable functions with docstrings, not just notebook cells, so it can be imported by teammates
- The **.ipynb notebook** itself, for interactive exploration and the validation sanity check
- Generated **dataset CSV(s)** actually used, so results are reproducible

Still Missing / To Add As the Project Progresses

- Rupsaa's neural-network repo (pipeline, training script, saved model, metrics)
- Raya's IMC mapping repo (estimation code, documented assumptions, final metrics)
- The common results table (`method`, `mae`, `rmse`, `r2`, `latency_ms`, `throughput`, `data_movement_bytes`, `energy_estimate`) populated with everyone's numbers
- Final comparison plots: accuracy, latency/speedup, data movement, energy, throughput
- A top-level project README tying all four components together into one pipeline story
- Demo video (or a link/script for it) ahead of the Sep 18 deadline

Generated to accompany the Core 4 hackathon project plan and one-page proposal.